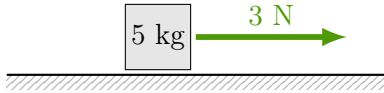


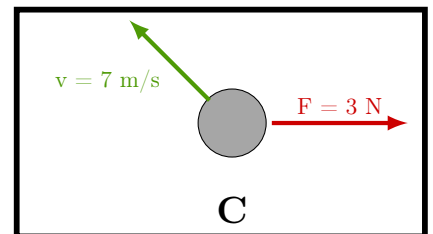
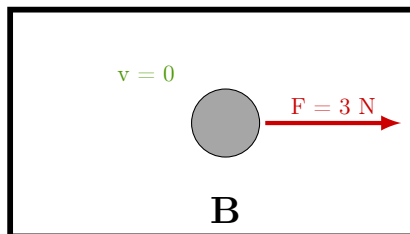
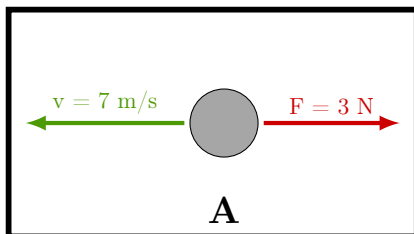
HW 1.5a

Name(s): _____

1. A 5 kg block has a force applied to it. Determine the acceleration of the block

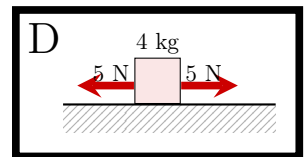
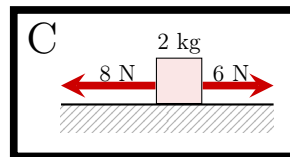
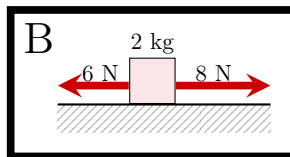
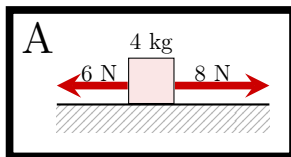


2. Three identical hockey pucks are on the ice.



Rank the pucks according to the size of their accelerations.

3. Two forces act on a block that is initially at rest

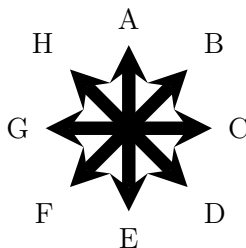
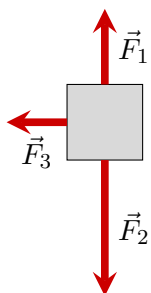


Rank the size of the accelerations of the the block from least to greatest

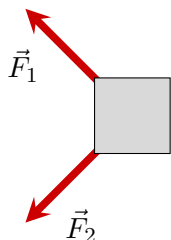
4. Gravity is a force that _____.

- A. attracts all metals in the Universe to all other metals in the Universe
- B. repels all metals in the Universe from all other metals in the Universe
- C. attracts all matter in the Universe to all other matter in the Universe
- D. repels all matter in the Universe from all other matter in the Universe

5. Three forces act on an object. In which direction does the object accelerate?

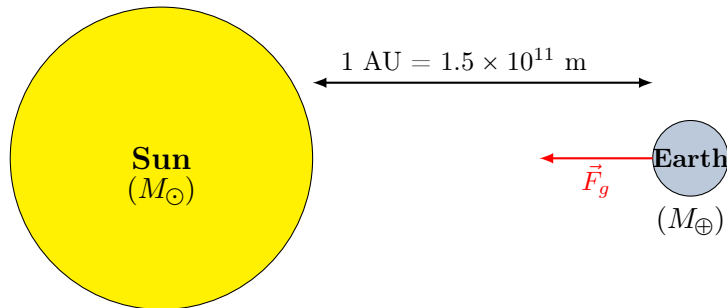


6. Three forces act on an object, but the object does not accelerate. Only two forces are shown. Draw the missing force, $\vec{F}_3$.

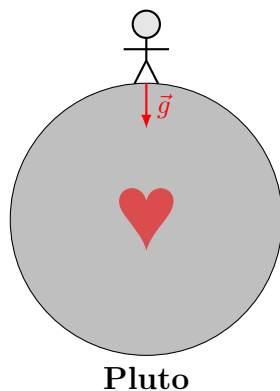


7. A constant force is applied to an object, causing the object to accelerate at 10 m/s^2 . What will the acceleration be if
- A. The force is halved?
 - B. The object's mass is halved?
 - C. The force and the object's mass are both halved?
 - D. The force is halved and the objects mass is doubled?
8. Why doesn't the Moon fall down to into the Earth? It keeps a stable orbit because _____.
- A. it is moving fast enough to miss the Earth even though it is always accelerating towards it.
 - B. the magnetic force between the Earth and the Moon resists their being pushed together by gravity
 - C. it is supported by the Earth's atmosphere
 - D. the density of the Moon is lower than the density of the Earth

9. Calculate the force of gravity of the Sun on the Earth.



10. An astronaut ($m = 150$ kg) is shown standing on the dwarf planet Pluto. Pluto has a planetary radius of $R_{\text{pluto}} = 1,188$ km and has a mass of $M_{\text{pluto}} = 1.31 \times 10^{22}$ kg.
- A. Find the acceleration due to gravity, g on Pluto.
- B. What is the Astronaut's weight on Pluto?



11. Rank the gravitational force experienced by an Astronaut, standing on Planets of varying mass and radius (given in units relative to Earth).

Planet Earth: (M,R) = (1,1)

Planet Flump: (M,R) = (2,1)

Planet Grump: (M,R) = (2,2)

Planet Hrumpf: (M,R) = (1/2,1/2)

12. If the moon were made twice as massive, and moved three times further away, the gravitational force of the Earth on the Moon would become _____ times as large.

13. If you drop a feather and a hammer from the same height, the hammer will reach the ground first. This is because _____.
14. If you drop a feather and a hammer from the same height on the moon which will hit the ground first?
15. Imagine a spacecraft placed somewhere along a line between the Earth and the Sun. Is there a position along the line where the gravitational attraction of the Earth is balanced by the gravitational attraction of the Sun, each pointing opposite to each other?
- A. Yes. It occurs closer to the Sun than the Earth.
 B. Yes. It occurs closer to the Earth than the Sun.
 C. Yes. It occurs at the half-way point between the Earth and Sun.
 D. No. At no point along the line do these forces balance.
16. Suppose the force of gravity between two objects is 70 Newtons (a Newton is a measure of force). If the mass of one of the objects doubles, the force of gravity will become _____ Newtons.
17. The table below shows the masses and distances (expressed in solar Masses) between four different pairs of stars (Cases A – D).

Case	Mass of star #1	Distance between star #1 and star #2	Mass of star #2
A	4	2	2
B	2	2	8
C	8	4	4
D	1	3	5

Rank (from greatest to least) the strength of the gravitational force exerted between the pairs of stars in cases A - D.

Ranking Order: Greatest 1 _____ 2 _____ 3 _____ 4 _____ Least